EVERY LOCAL SUBRING OF A FIELD IS DOMINATED BY A VALUATION RING WITH THAT FRACTION FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted proof refers to no label; the Zorn argument and both enlargement cases are written out inside the span, so the card carries no navigation stub and no bibliography.

Allowed standard prerequisites (editorial).

- Zorn's lemma on a partially ordered set of local subrings ordered by domination. Used in the counted proof at lines 11974-11979. Produces a maximal element of the collection, which is then shown to be a valuation ring.
- Adjoining an element to a local subring. Used in the counted proof at lines 11984-11996. Constructs a strictly larger local subring dominating the given one whenever the fraction field is not yet everything.

Source reference. Original lemma 00GQ: sources/stacks-algebra/pinned-original-source.tex, source lines 7952–7952. Navigation pointer for \ref{lemma-integral-overring-surjective} used inside the counted span (line 11991); the argument that consumes it is retained in the span.

Lemma 0.1. *Let K be a field. Let $A \subset K$ be a local subring. Then there exists a valuation ring with fraction field K dominating A.*

Proof. We consider the collection of local subrings of K as a partially ordered set using the relation of domination. Suppose that $\{A_i\}_{i \in I}$ is a totally ordered collection of local subrings of K. Then $B = \bigcup A_i$ is a local subring which dominates all of the A_i. Hence by Zorn's Lemma, it suffices to show that if $A \subset K$ is a local

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ring whose fraction field is not K, then there exists a local ring $B \subset K$, $B \neq A$ dominating A.

Pick $t \in K$ which is not in the fraction field of A. If t is transcendental over A, then $A[t] \subset K$ and hence $A[t]_{(t,\mathfrak{m})} \subset K$ is a local ring distinct from A dominating A. Suppose t is algebraic over A. Then for some nonzero $a \in A$ the element at is integral over A. In this case the subring $A' \subset K$ generated by A and ta is finite over A. By Lemma 00GQ there exists a prime ideal $\mathfrak{m}' \subset A'$ lying over $\mathfrak{m}$. Then $A'_{\mathfrak{m}'}$ dominates A. If $A = A'_{\mathfrak{m}'}$, then t is in the fraction field of A which we assumed not to be the case. Thus $A \neq A'_{\mathfrak{m}'}$ as desired. $\square$